COMPUTER GRAPHICS CO 303

TOTAL MARKS: 20

TIME: 45min

Questions 1 to 5 carries 1 mark each.

1. In random scan display, the frame buffer holds ....
   a) Line drawing commands b) Scanning instructions
   c) Image Resolution d) Intensity information

2. Brightness of a display is controlled by varying the voltage on the ....
   a) Focusing anode b) Connection pins c) Control grid d) Power supply

3. The purpose of refreshing a CRT is ....
   a) To avoid flickering b) To maintain steady picture
   c) To avoid fading of pixels d) All of the above

4. The screen resolution is 1024x 1024 and the frame-buffer address of (0,0) is 6028. The address of the pixel (60, 999) is ______________.

5. Which is an example of non-emissive displays
   a) LED b) LCD c) Gas Discharge tube d) Plasma Panel

6. Consider a graphics system with a resolution of 64 x 64 pixels. The color look-up table has 16 entries. Calculate the storage requirement for direct and indirect method. Colors are represented by 24 bits. [3+3=6]

7. Explain the basic refresh operation of the video controller. [3]

8. Use the Scanline Polyfill algorithm to fill the polygon ABCDEA. The coordinates of the vertices are A(0,0),B(4,3), C(8,0), D(8,8),E(3,8). Show the SET and AET structures for each step of your algorithm. [6]